

Overview

1 Monte Carlo integration

- Recap
- Error estimates
- Importance sampling

2 Monte Carlo methods in statistical physics

3 Summary

Quiz next week!

Monte Carlo integration

Basic idea

Integrate functions by randomly sampling points

$$\int_V f(x) dx = V \langle f \rangle \approx \frac{V}{N} \sum_{i=1}^N f(x_i)$$

where x_i are randomly distributed over V .

Error estimates

$$\sigma^2 = \frac{V^2}{N} \left[\langle f^2 \rangle - \langle f \rangle^2 \right] \approx \frac{V^2}{N} \left[\frac{1}{N} \sum_{i=1}^N f(x_i)^2 - \left(\frac{1}{N} \sum_{i=1}^N f(x_i) \right)^2 \right]$$

- The error decreases as $1/\sqrt{N}$
- Superior to fixed-grid integration in high dimensions

Monte Carlo integration

MC integration in a nutshell

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Advantages

- It works in arbitrary numbers of dimensions: $\int f dV = V \langle f \rangle$
- It works for complicated boundaries
- It is superior to fixed grid in higher dimensions (next!)

Error estimates

The variance of a stochastic variable X is

$$\sigma_X^2 \equiv \text{var } X \equiv \langle (X - \langle X \rangle)^2 \rangle = \langle X^2 \rangle - \langle X \rangle^2$$

For our MC integral we get

$$\begin{aligned}\sigma^2 &= \left\langle \left(\frac{V}{N} \sum_i f_i \right)^2 \right\rangle - \left\langle \frac{V}{N} \sum_i f_i \right\rangle^2 \\&= \frac{V^2}{N} \left[\frac{1}{N} \left\langle \sum_i f_i^2 \right\rangle + \frac{1}{N} \left\langle \sum_{i \neq j} f_i f_j \right\rangle - N \langle f \rangle^2 \right] \\&= \frac{V^2}{N} \left(\langle f^2 \rangle + \frac{1}{N} \sum_{i \neq j} \langle f_i \rangle \langle f_j \rangle - N \langle f \rangle^2 \right) \\&= \frac{V^2}{N} \left(\langle f^2 \rangle + \frac{N(N-1)}{N} \langle f \rangle^2 - N \langle f \rangle^2 \right) = \frac{V^2}{N} \left(\langle f^2 \rangle - \langle f \rangle^2 \right)\end{aligned}$$

In going from the second to the third line we have used that f_i and f_j are independent and uncorrelated.

Advantages in many dimensions

Standard numerical integration gives errors $\propto$ powers of the grid spacing,

$$\text{Trapezium: } I_T - I \sim \delta^2, \quad \text{Simpson: } I_S - I \sim \delta^3, \dots$$

In one dimension we have $\delta \propto 1/N$, so adding more points gives much faster improvement in accuracy than MC:

Trapezium:	$N \rightarrow 2N$	$\implies$	$\text{err} \rightarrow \text{err}/4$
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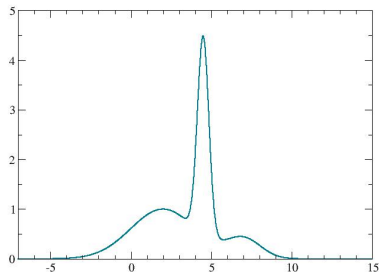
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So for trapezium ($k = 2$) Monte Carlo is better once $d > 4$

Importance sampling

Assume we have a function that is sharply peaked in some region, and close to zero in others

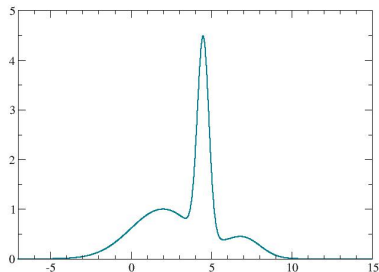
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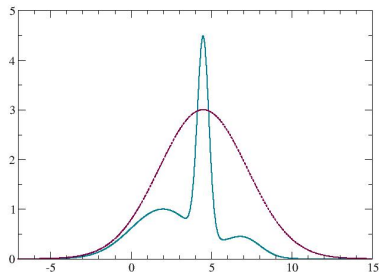


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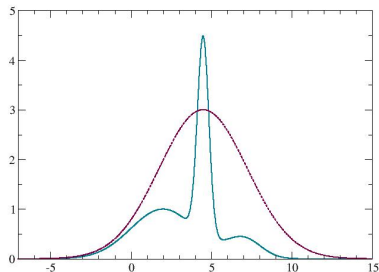
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$$I = \int f(x) dx = \int q(x) \left[\frac{f(x)}{q(x)} \right] dx \approx \frac{1}{N} \sum_{i=1}^N \frac{f(x_i)}{q(x_i)}$$

Variance is much reduced!

Statistical physics

Microscopically

Large number of individual degrees of freedom $\{p_i, q_i\}$, $H(p, q)$

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A few thermodynamical control variables: T, μ, V

and bulk thermodynamical properties: $\varepsilon, P, S, \dots$

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Statistical physics: bridge micro—macro

Work with ensembles of microscopic configurations

Probability of configuration $X = \{p_i, q_i\}$ is

$$P(X) \propto e^{-\beta H(X)} \equiv e^{-H(X)/k_B T}$$

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Common recipe

Construct a series of configurations via a stochastic process — next!

Summary

Monte Carlo integration

- Errors decrease as $1/\sqrt{N}$ regardless of dimensionality
- Superior to standard integration in higher dimensions.
- Importance sampling can drastically improve performance